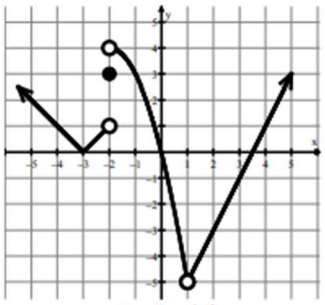
Worksheet 1-5 B
Limits of Piecewise and Composite Functions #1
AP Calculus

Name_____

BE SURE TO SHOW ALL WORK.

Use the graph to find the given limits

1.)

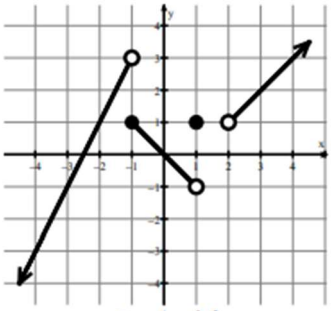


Graph of f

a.) $\lim_{x\to 3} f(f(x))$

b.) $\lim_{x\to 1} f(f(x))$

2.)

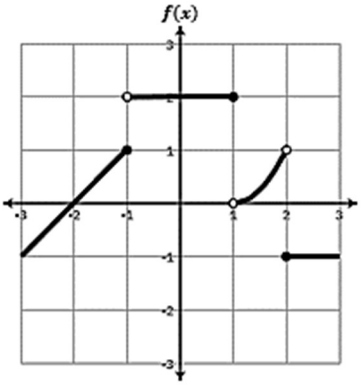


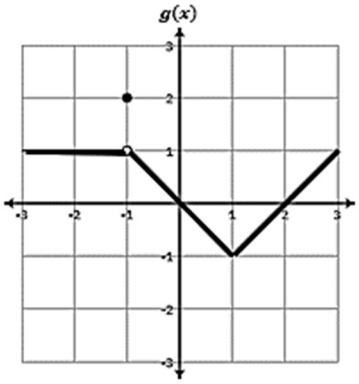
Graph of f

a.) $\lim_{x\to -2} f(f(x))$

b.) $\lim_{x\to 4} f(f(x))$

Using the graphs of $f(x)$ and $g(x)$ provided at the right, determine each of the following limits or state that they do not exist. Circle your final answers





3. $\lim_{x\to -1^+} f(x)$	**4.** $\lim_{x\to -1^-} f(x)$	**5.** $\lim_{x\to -1} f(x)$
6. $\lim_{x\to -1^+} g(x)$	**7.** $\lim_{x\to -1^-} g(x)$	**8.** $\lim_{x\to -1} g(x)$
9. $g(-1)$	**10.** $\lim_{x\to 0} \frac{f(x)}{g(x)}$	

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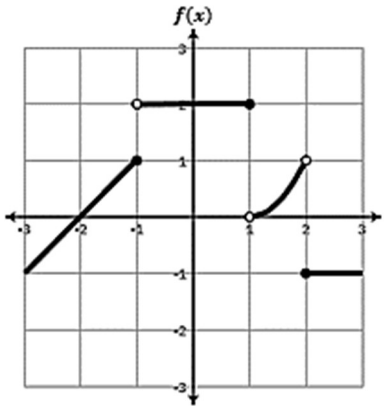

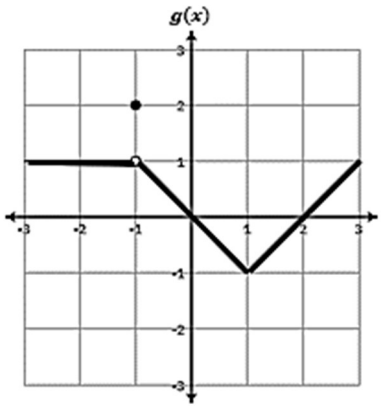

11. $\lim\limits_{x\to 0}\dfrac{g(x)}{f(x)}$	**12.** $\lim\limits_{x\to -1}\left[f(x)\cdot g(x)\right]$	**13.** $\lim\limits_{x\to 2}\left[f(x)\cdot g(x)\right]$
14. $\lim\limits_{x\to 0}\left[2f(x)+3g(x)\right]$	**15.** $\lim\limits_{x\to 0^-} f(x+2)$	**16.** $\lim\limits_{x\to -1^-} f(x^2)$
17. $\lim\limits_{x\to -2} g\left(f(x)\right)$	**18.** $\lim\limits_{x\to 2^-} f\left(f(x)\right)$	**19.** $\lim\limits_{x\to 2^+} f\left(-f(x)\right)$
20. $\lim\limits_{x\to -1} f\left(g(x)-2.5\right)$	**21.** $\lim\limits_{x\to 0} f\left(\left[g(x)\right]^2+1\right)$	

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